A LOW-CODE, NO-CODE
INFORMATICS PLATFORM

Sapio AI

Sapio now brings the fast paced, evolving world of Artificial Intelligence (AI) to your life science enterprises and cutting edge research. Tackle research obstacles with two new AI features: ELaiN and Sapio Support Agent.

ELaiN (Electronic Laboratory AI Notebook) is a chat feature where users can ask the AI to create new experiments, find and compare data points, and run inventory or reagent analysis.

The Sapio Support Agent is a chat feature where users can ask the AI questions about how to use their platform or to generate Python code. The AI will return explanations and instructions based on Sapio technical documentation.

This guide will review ELaiN and Sapio Support Agent features and common use cases. Read the entire guide or skip to your section of interest.

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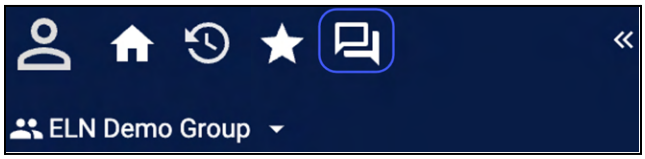
ELaiN

ELaiN is the Electronic Laboratory AI Notebook enhancement that can be used to design or create experiments and evaluate existing data points or inventory upon prompt. Prompts can be questions or commands. See the following sections for how to access and use ELaiN.

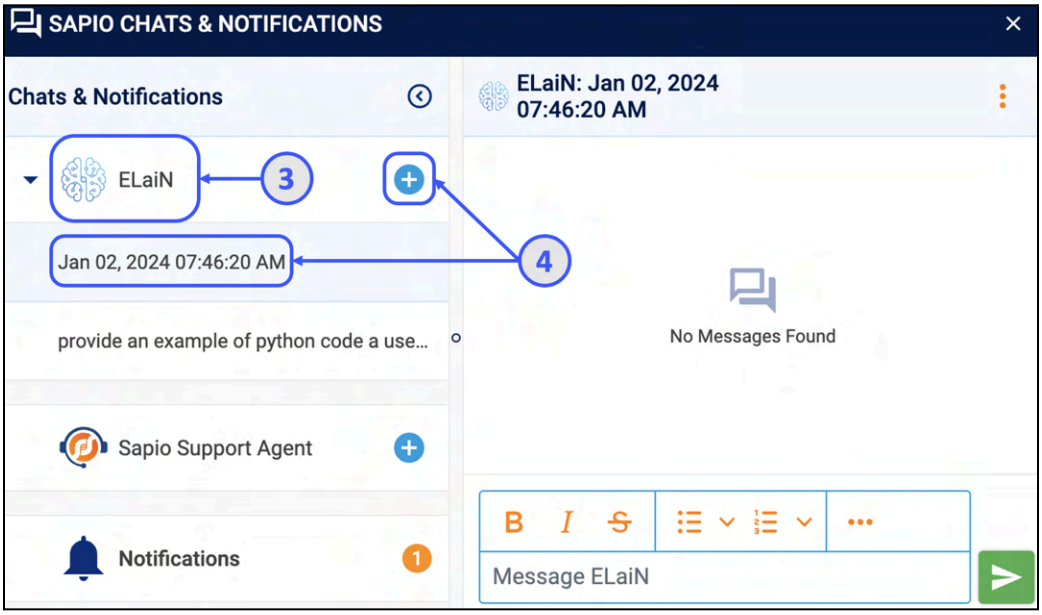
Navigate to ELaiN

ELaiN can be accessed from a user's Home Page in Sapio LIMS, ELN, or Jarvis.

1. Login to Sapio
2. Click the message box icon to open the Chats and Notifications widget



3. Choose the 'ELaiN' banner
4. Select an existing conversation or click the + button to start a new conversation



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Use ELaiN to Create a New Experiment

Design and create new experiments from existing templates using ELaiN. From the ELaiN chat window…

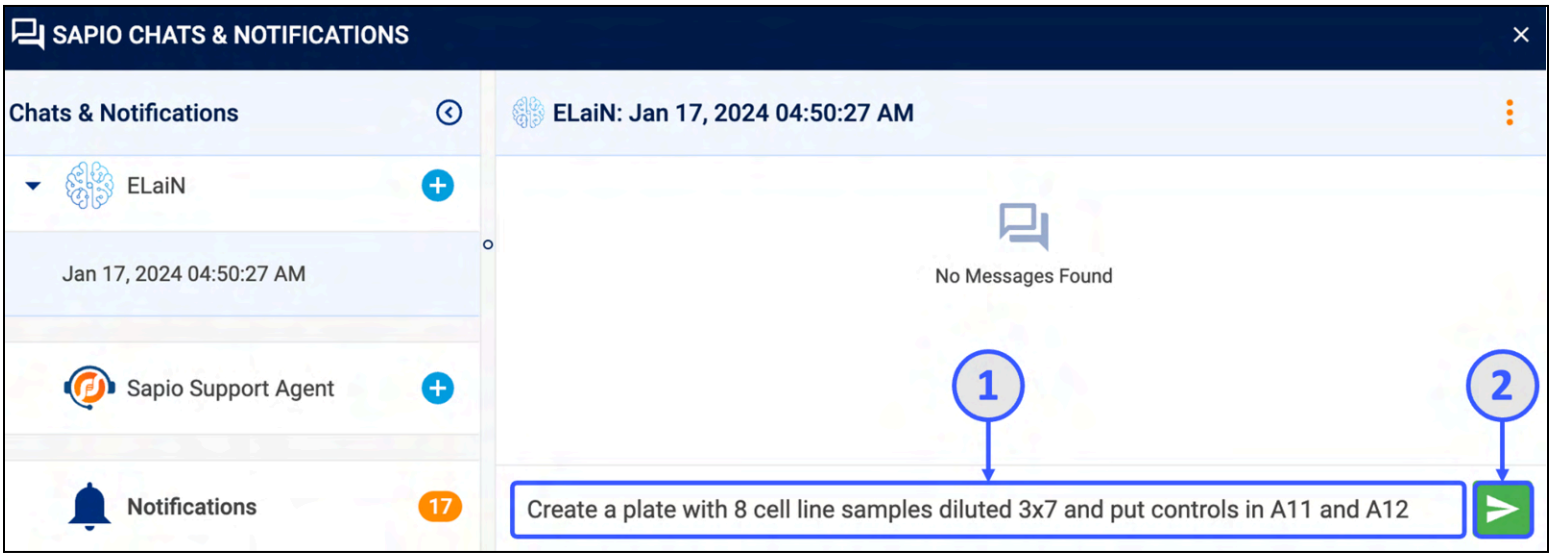
1. Type a message in the 'Message ELaiN' box

Tip : Example prompts such as…

- *"Create a plate with 8 cell line samples diluted 3x7 and put controls in A11 and A12,"*
- *"Design a new experiment using the PCR template"*
- *and "Add viral vector samples to the experiment"*

…will result in the creation of a new experiment with relevant sections. Generated sections in the new experiment (instrument tracking, samples, setup plate, or reagent tracking) depend on what is configured in your environment.

2. Click the send airplane button



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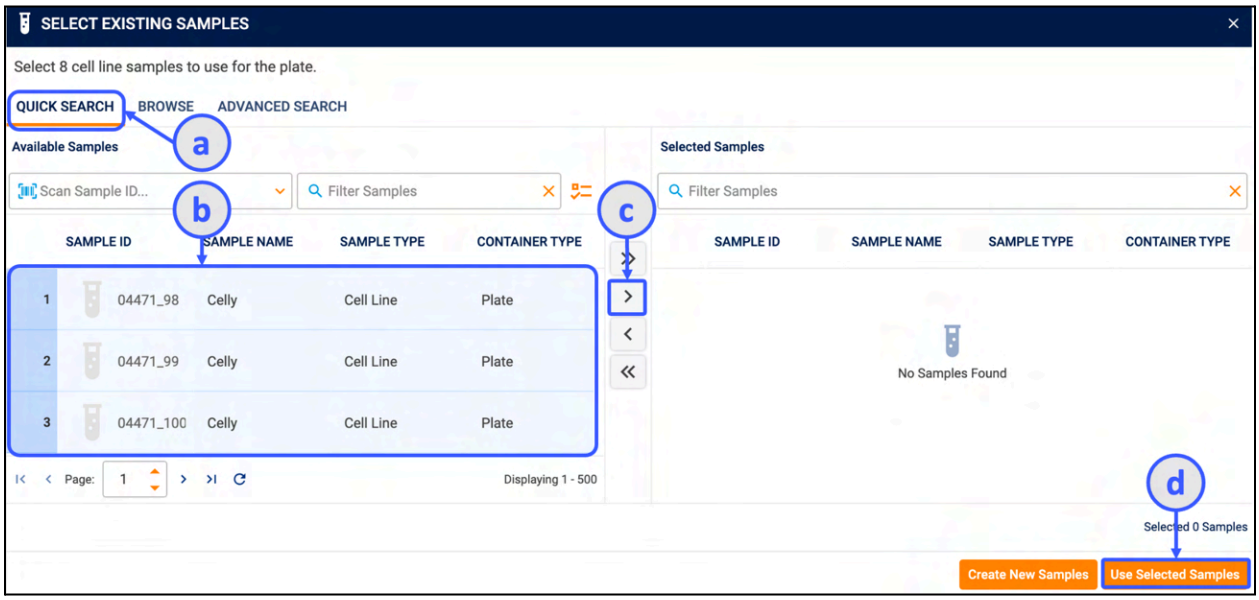
3. If prompted, select from the existing available samples in the list or create new samples

- **For Existing Samples :**

a. Select Quick Search, Browse, or Advanced Search
b. Choose the samples
c. Click the > arrow to move the available samples to the selected side
d. Then click 'Use Selected Samples'



- **For New Samples :** Click 'Create New Samples'

Tip : Samples can also be added by scanning Sample IDs or filtering available samples

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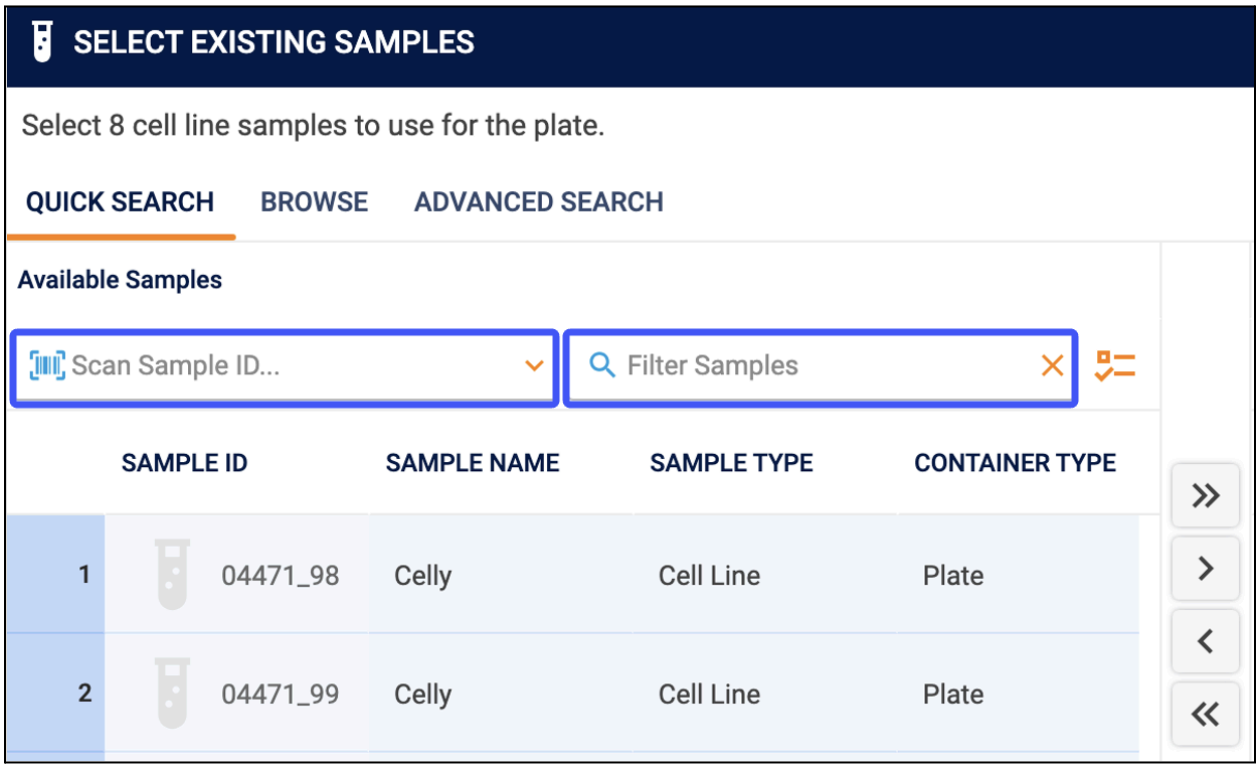

4. This will open a window with the new experiment. Fill in any missing details and proceed to conduct the experiment.

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SAPIO SEAMLESS ELN ®

1. Instrument Tracking
2. Samples
3. Setup Plate

Demo User's Experiment (7974917)

E-Sign | Cancel Experiment

Experiment Information

Created By: demo.user@sapiosciences.com

Date Created: Jan 03, 2024 12:18:42

PROCEDURE DETAILS

Instrument Tracking

	INSTRUMENT TYPE *	INSTRUMENT USED	TECHNICIAN	DATE	MODEL NUMBER	SERIAL NUMBER	DATE PUI
1	PCR						

Samples

	SAMPLE ID	SAMPLE NAME	SAMPLE TYPE	CONTAINER TYPE	QUANTITY	QUANTITY UNITS
1	04471_98	Celly	Cell Line	Plate		mL
2	04471_99	Celly	Cell Line	Plate		mL
3	04471_100	Celly	Cell Line	Plate		mL

Status: Active | EXP ID: 7974917

SAPIO SEAMLESS ELN ®

1. Instrument Tracking
2. Samples
3. Setup Plate

Demo User's Experiment (7974917)

E-Sign | Cancel Experiment

Experiment Information

Created By: demo.user@sapiosciences.com

Date Created: Jan 03, 2024 12:18:42

PROCEDURE DETAILS

Setup Plate

AS

Samples | ADD+

Plate ID: 3569 | ADD PLATE+

	SAMPLE ID	SAMPLE NAI
1	52194	
2	52195	
3	52196	
4	52197	
5	52198	
6	52199	

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Use ELaiN to Review or Compare Inventory/Reagents

Review or compare inventory:reagents using ELaiN. From the ELaiN chat window…

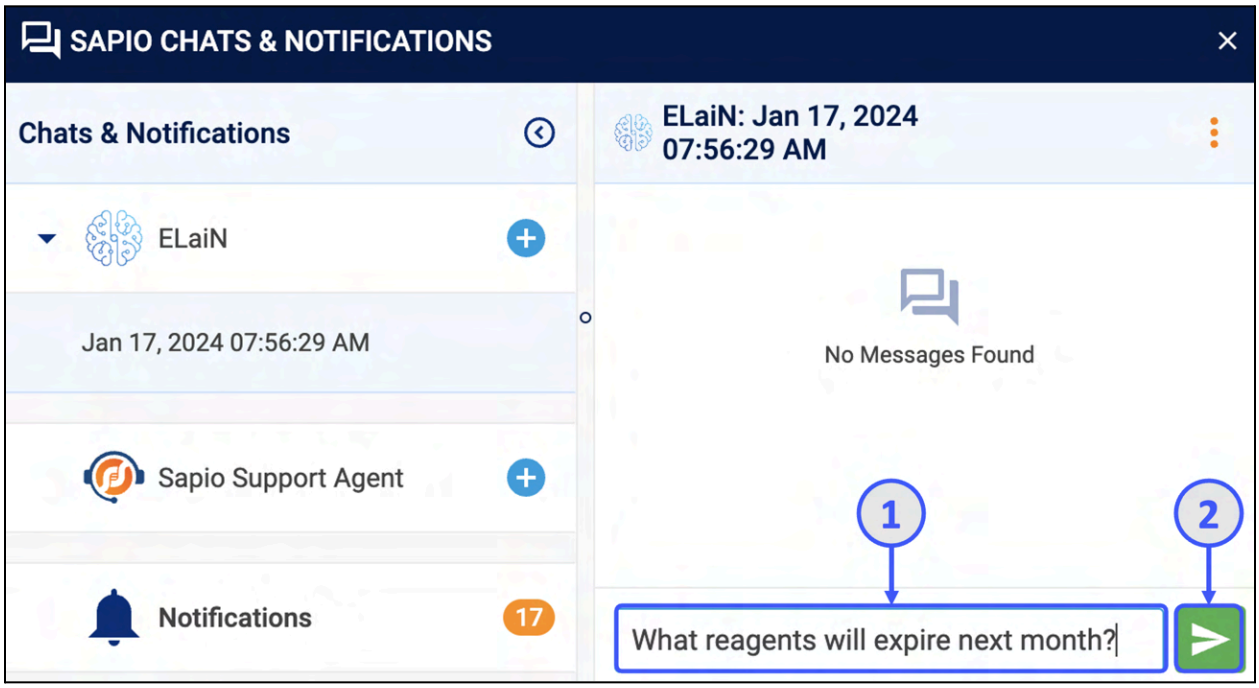
1. Type a message in the 'Message ELaiN' box

Tip : Example prompts such as…

- *"Find me all the DNA samples I created this year"*
- *"What reagents will expire next month?"*

…will result in the creation of a data table list with the relevant results. Generated lists display as the result of an Advanced Search that ELaiN has run on your behalf to address the question.

2. Click the send airplane button



3. This will open a window with the relevant results listed in a data table. The generated data table is displayed as the results of an Advanced Search that ELaiN has run on your behalf. Filter, create a chart, export a CSV of results.

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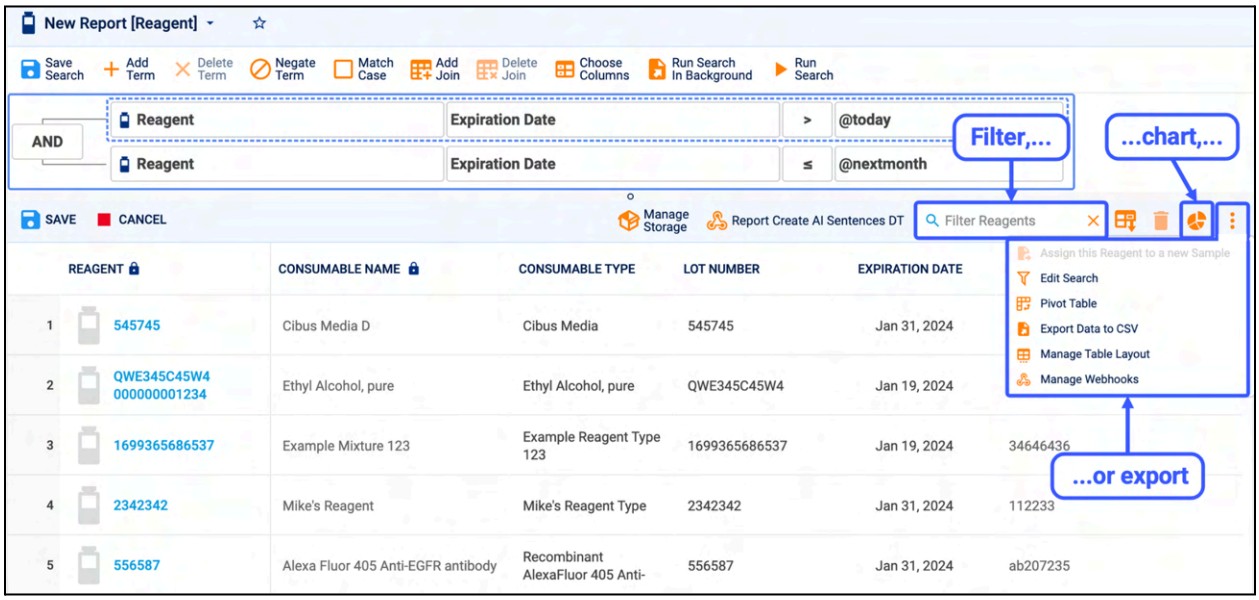
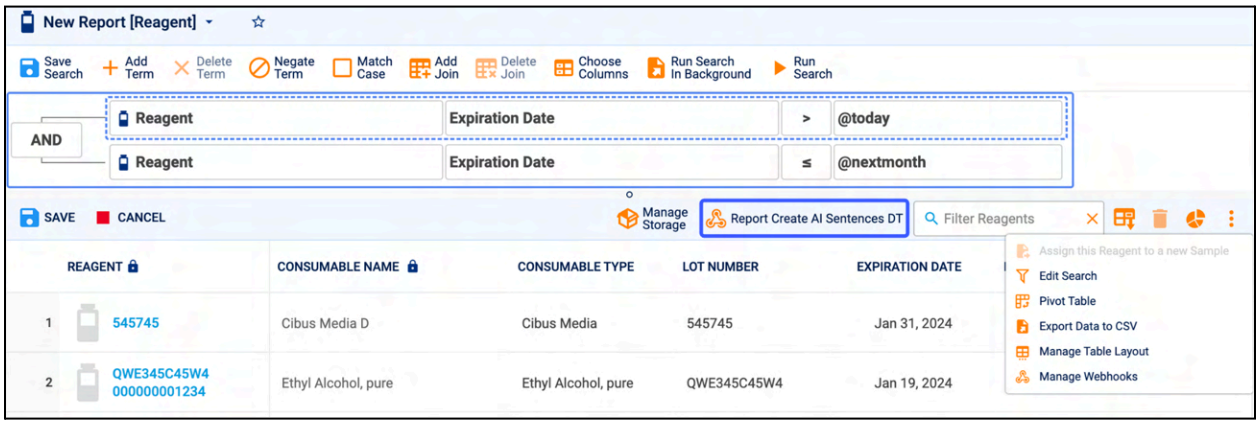


Tip : Click the 'Report Create AI Sentences DT' button to see a list of other possible ai prompts that would have yielded the same results as the current displayed data table



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GPT RUN RESULT

Response :

PROMPT:
Give me all of the reagents that are set to expire within the next month.

RESULTING JSON:
{"termType": "COMPOSITE_TERM", "negated": false, "termOperation": "AND_OPERATOR", "leftChild": {"termType": "RAW_TERM", "negated": false, "dataTypeName": "ConsumableItem", "dataFieldName": "ExpirationDate", "termOperation": "LESS_THAN_OR_EQUAL_OPERATOR", "trim": false, "value": "@nextmonth"}, "rightChild": {"termType": "RAW_TERM", "negated": false, "dataTypeName": "ConsumableItem", "dataFieldName": "ExpirationDate", "termOperation": "GREATER_THAN_OPERATOR", "trim": false, "value": "@today"}}

PROMPT:
Can you show me the reagents that will expire within the next month?

RESULTING JSON:
{"termType": "COMPOSITE_TERM", "negated": false, "termOperation": "AND_OPERATOR", "leftChild": {"termType": "RAW_TERM", "negated": false, "dataTypeName": "ConsumableItem", "dataFieldName": "ExpirationDate", "termOperation": "LESS_THAN_OR_EQUAL_OPERATOR", "trim": false, "value": "@nextmonth"}, "rightChild": {"termType": "RAW_TERM", "negated": false, "dataTypeName": "ConsumableItem", "dataFieldName": "ExpirationDate", "termOperation": "GREATER_THAN_OPERATOR", "trim": false, "value": "@today"}}

PROMPT:
I would like to see all the reagents with an expiration date within the next month

OK

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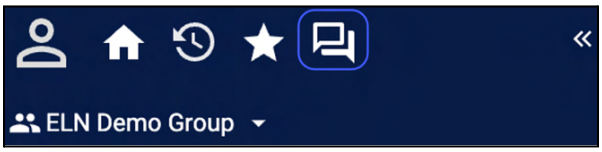
Sapio Support Agent

The Sapio Support Agent is an enhancement that can be used to ask questions and get answers about the platform or to generate python code for drafting connections to your system through prompts. Prompts can be questions or commands. The AI will return explanations, instructions, or code based on Sapio technical documentation.

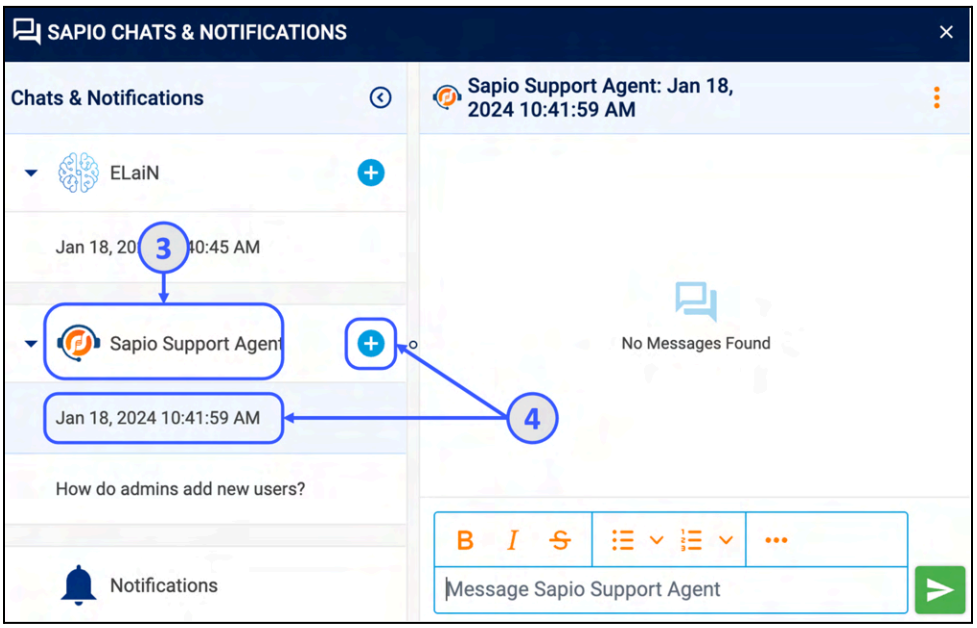
Navigate to Sapio Support Agent

The Sapio Support Agent can be accessed from a user's Home Page in Sapio LIMS, ELN, or Jarvis.

1. Login to Sapio
2. Click the message box icon to open the Chats and Notifications widget



3. Choose the 'Sapio Support Agent' banner
4. Select an existing conversation or click the + button to start a new conversation



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Ask Sapio Support Agent Product Use Questions

Get answers for how-to use Sapio Products with Sapio Support Agent. From the Sapio Support Agent chat window…

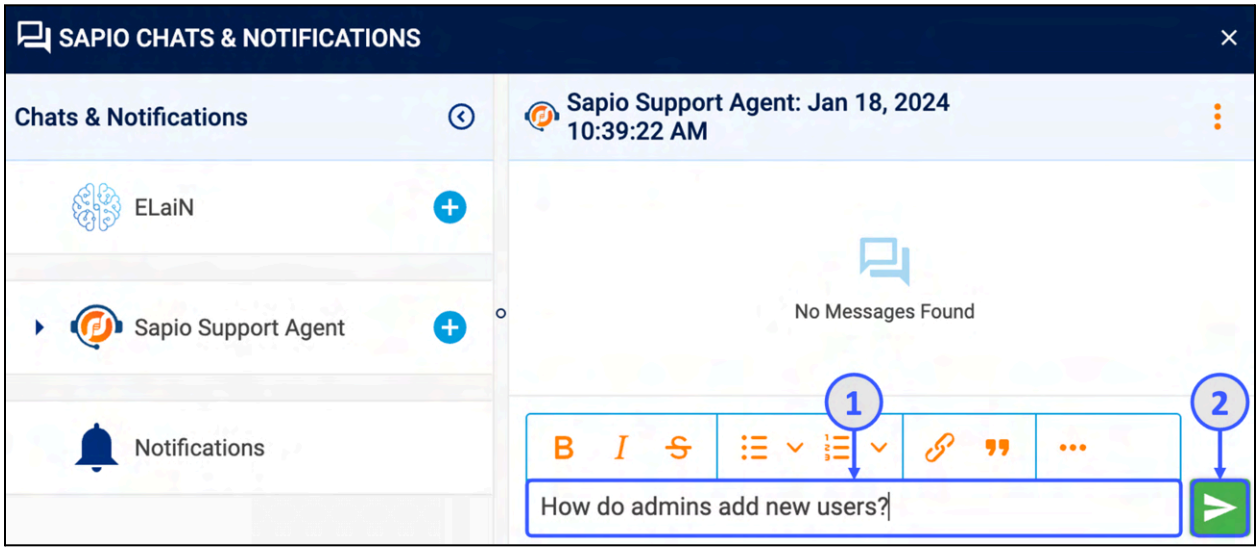
1. Type a message in the 'Message Sapio Support' box

Tip : Example prompts such as…

- *"How do Admins add Users?"*
- *"How can I remove users from Groups?"*
- *"What is a Data Type?"*

…will result in the generation of a response with step-by-step instructions or an explanation. Additional questions can be asked from there.

2. Click the send airplane button



3. This will open a window with step-by-step instructions or an explanation. The generated answers are rendered from Sapio documentation that the AI has consumed. Additional questions can be asked or a new chat started.

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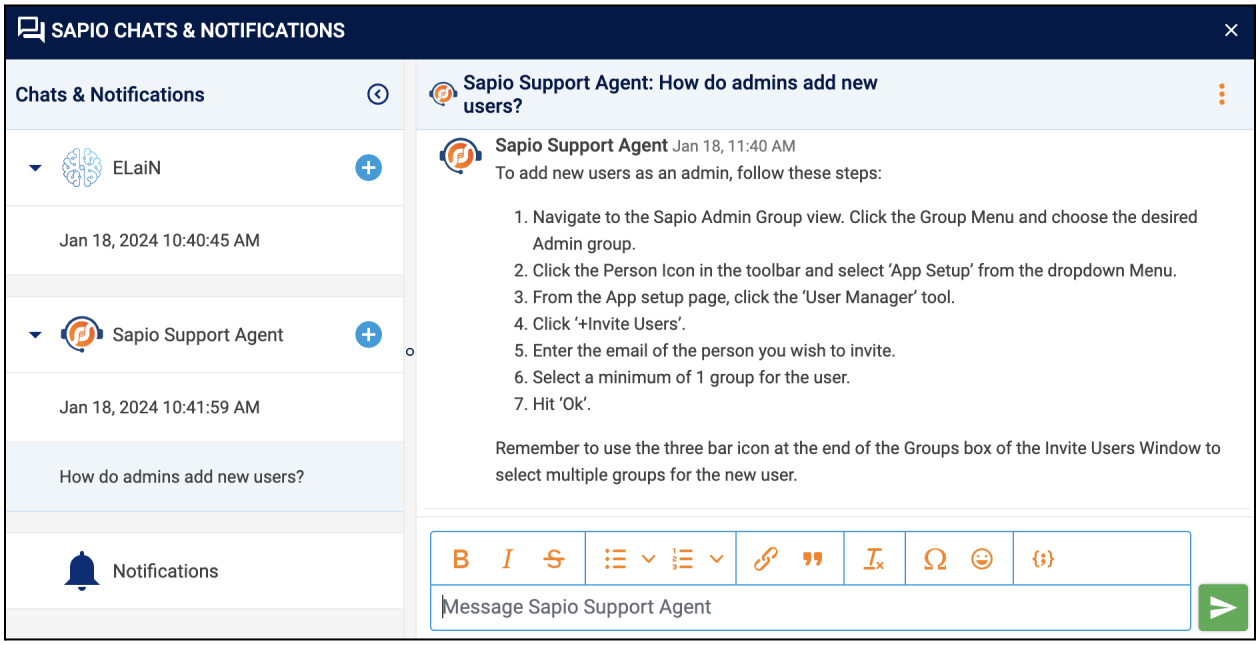

Use Sapio Support Agent to Create Python Code

Get use cases and examples of python code applications in Sapio Products with Sapio Support Agent. From the Sapio Support Agent chat window…

1. Type a message in the 'Message Sapio Support' box

Tip : Example prompts such as…

- *"How can I create an experiment called 'Test Experiment' that is based on the 'Test' Template in python code?"*
- *"Provide an example of coding"*
- *"What can I do with code in Sapio"*

…will result in the generation of a response with code examples and explanations. Additional questions can be asked from there.

2. Click the send airplane button

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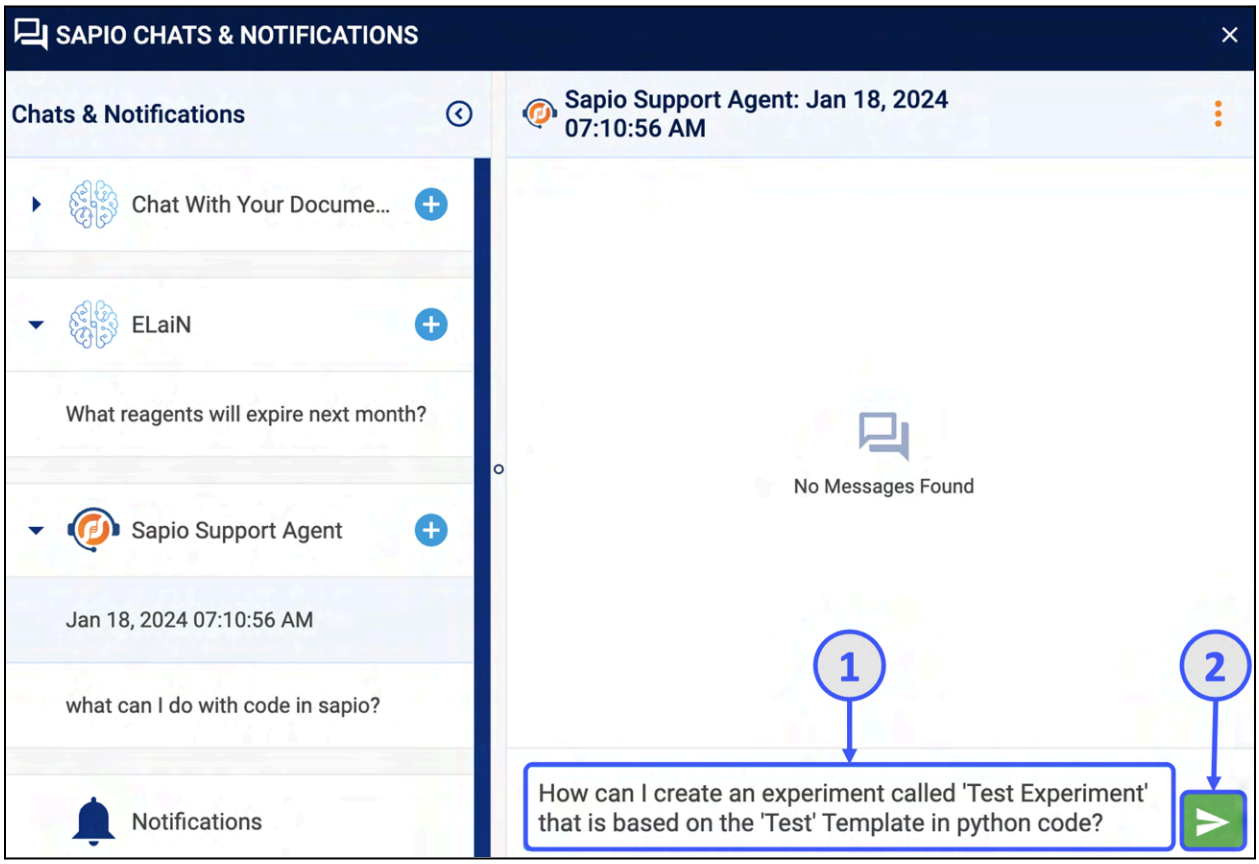
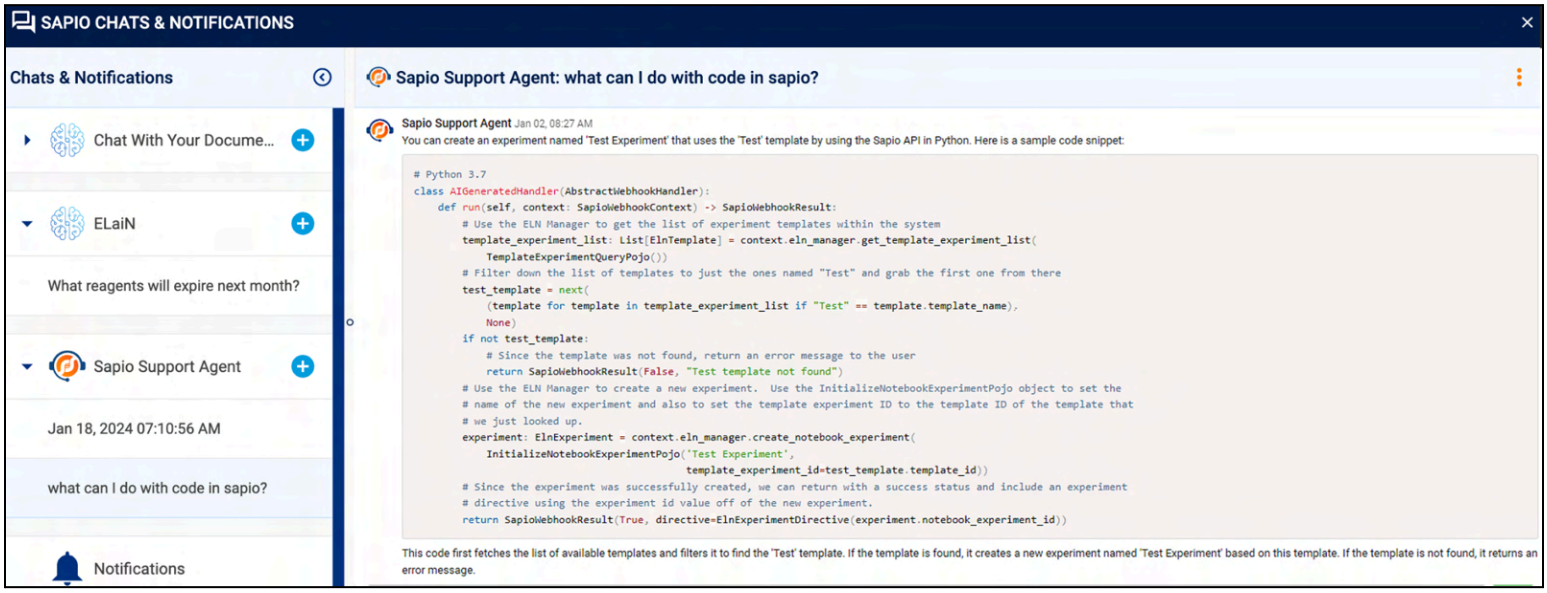


3. This will open a window with explanations and example code. The generated answers are rendered from Sapio documentation that the AI has consumed. Additional questions can be asked or a new chat started.



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